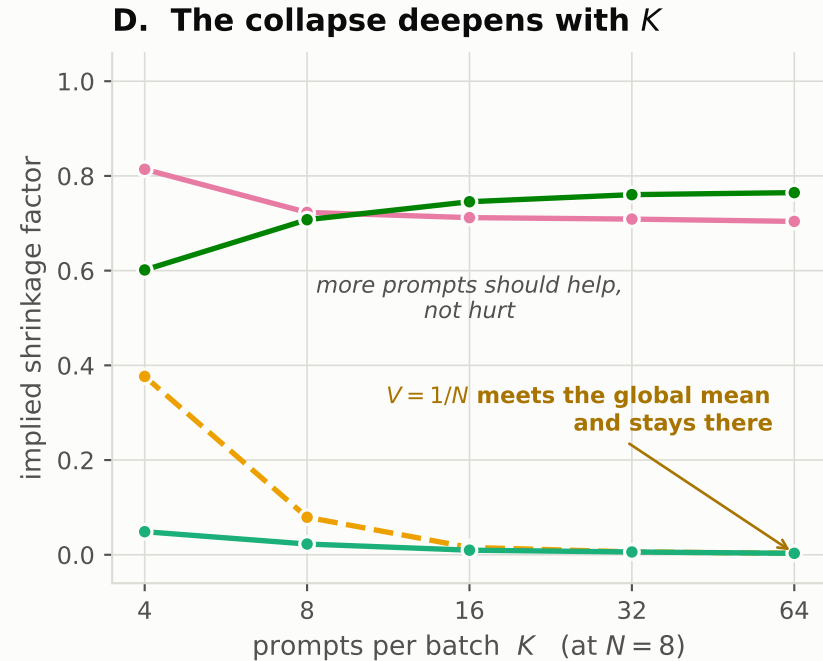
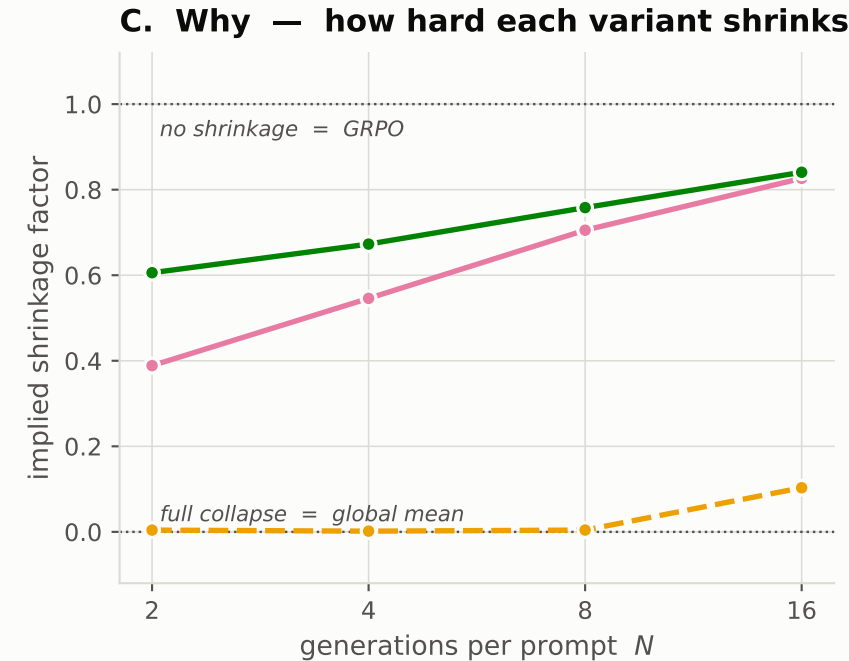
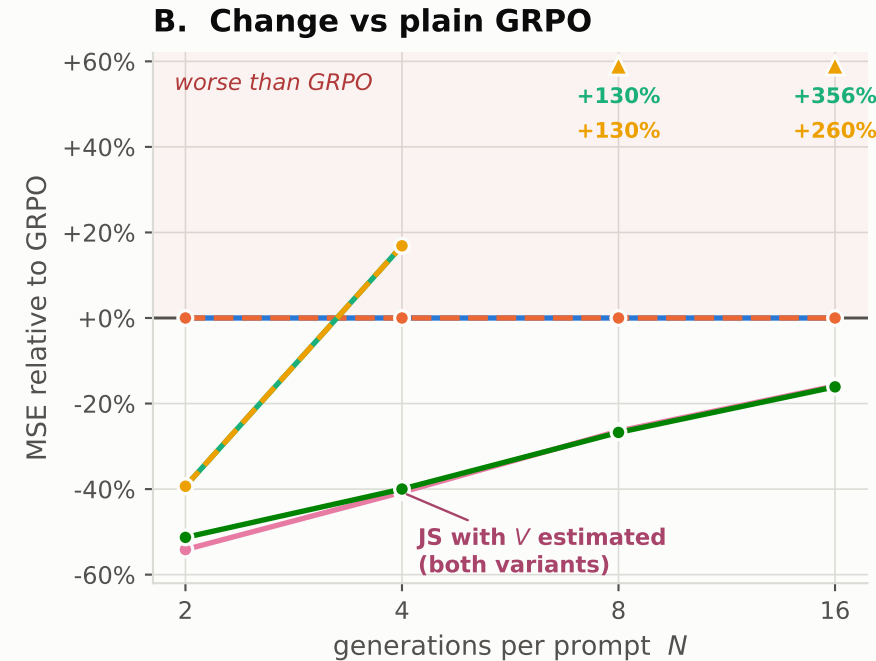
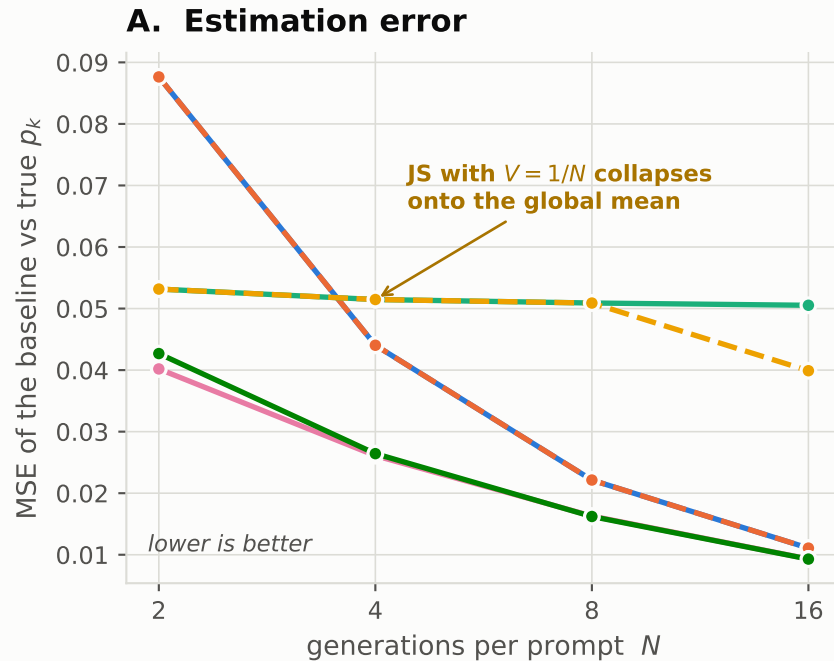


**Shrinkage baselines for GRPO — Bernoulli rewards,  $p_k \sim \text{Beta}(1.2, 2.2)$ . A-C:  $K=32$ , sweeping  $N$ . D:  $N=8$ , sweeping  $K$ .**



GRPO (group mean) RLOO (leave-one-out) global mean JS,  $V=1/N$  (GRPO.md as written) JS,  $V$  estimated JS two-level LOO (arXiv:2511.03710)